

Arna Ghosh

Integrated Program in Neuroscience, McGill University, Montréal

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EDUCATION



MSC. NEUROSCIENCE

McGill University, Montréal, Canada

Sep 2017 - Present

CGPA: 4.0/4.0



BACHELOR OF TECHNOLOGY

Indian Institute of Technology Kharagpur, India

Jul 2013 - May 2017

CGPA: 9.26/10

Majors in Electrical Engineering, Minors in Computer Science

Micro-specialization in Embedded Wireless Technology

Awarded the **Silver medal** for achieving highest CGPA among graduating students of department

INTERNSHIPS



MCGILL UNIVERSITY, MITACS Globalink Research intern

May 2016 - Jul 2016

Mechanism underlying sensorimotor integration during motor learning

Dr. Marie-Hélène Boudrias

- Designed the software and hardware interface for tasks to study motor learning characteristics under different external stimulation conditions using **Python**, **C++** and **OpenCV**
- Interfaced EEG, EMG and fMRI setups to the designed software for different experiments. [Media Coverage]



UNIVERSITY OF BRITISH COLUMBIA OKANAGAN, Summer intern

May 2015 - Jul 2015

Slide-Scanning Microscopy on a Smart Phone by High-Frame-Rate Video Capture

Dr. Kenneth Chau

- Developed the image processing module for fast deblurring of microscopic images extracted from video and developed an integrated hardware-software interface using an Arduino
- Modified **OpenCV** library to allow 240 fps video capture on iOS using **Objective C**.

MAJOR PROJECTS



Deep Learning for Neuroimaging data analysis, Masters Thesis Project

Sep 2017 - Present

Dr. Marie-Hélène Boudrias & Dr. Georgios Mitsis

- Working on a DL framework to identify discriminative “biomarkers” for groups (treatment vs control) from EEG. Developed **ccCAM** (cue-combination for Class Activation Map), a technique inspired from human perception studies, to improve network interpretability
- Working on a ML- and DL-driven **brain age prediction** system from MRI and MEG recordings. The system will be used to predict capability of stroke recovery through rehabilitation therapeutic techniques
- Designed an experiment using GANs for faces to study human “**memory**” cells from fMRI and iEEG recordings



Brexting: Brain Texting, Hobby Project

Dec 2017 - Aug 2018

Intel Innovate FPGA Grand Finalists

Built a DL framework on an Intel FPGA board for decoding of imagined movements from EEG recordings to enable real-time typing. Won the **Silver award** in North Americas regional finals and **Iron award** in grand finals Repo



Deep Learning for mitotic figure detection, B.Tech Project

Jul 2016 - May 2017

Dr. Debdoot Sheet

Built a DL framework to localize and identify mitotic nuclei from *breast histopathological* images on **Torch** and extended the idea to *leukemia detection*. Awarded the **Systems Society award** for best project work on Systems



Applications of Deep Learning in Computer Vision, Hobby project

Jul 2016 - Present

An attempt to solve computer vision problems using biologically plausible algorithms

- Implementing DCGAN for learning handwriting style for characters to improve handwriting style for words
- Extending text-to-image GAN to predict subsequent scene in a racing game given the present scene and user action
- Working on a simultaneous localization and classification framework for objects in a scene based on the importance of filters in CNNs with wise applicability to various domains



Dr. Pawan Goyal

Implemented a binarisation algorithm to threshold scanned images and segregate text from background to detect words in a document written in an Indian language. Funded by the **Google-IIT Pilot** program and accepted to be integrated into Google Translate



Dr. Debashish Chakravarty

- Developed image processing modules to detect obstacles, detect lanes and construct a world map by fusing data from camera and LIDAR on **Robot Operating System (ROS)**
- Worked on efficient road segmentation for off-road conditions and visual odometry problems using Convolutional Neural Networks

PUBLICATIONS

Conference Papers and Preprints:

1. **A. Ghosh**, F. Dal Maso, M. Roig, G.D. Mitsis, M.H. Boudrias, *Deep Semantic Architecture with discriminative feature visualization for neuroimage analysis*, Link
2. **A Ghosh**, S Singh, D Sheet, *Simultaneous Localization and Classification of Acute Lymphoblastic Leukemic Cells in Peripheral Blood Smears Using a Deep Convolutional Network with Average Pooling Layer*, IEEE 12th International Conference on Industrial and Information Systems (ICIIS), December 2017.
3. V Mohanty, S Agrawal, S Datta, **A Ghosh**, V D Sharma, D Chakravarty, *DeepVO: A Deep Learning approach for Monocular Visual Odometry*, Link

Poster Presentations:

1. **A. Ghosh**, F. Dal Maso, G.D. Mitsis, M.H. Boudrias, *Deep Convolutional Architecture with discriminative feature visualization for analysis of EEG data*, Neuroinformatics, August 2018.
2. **A. Ghosh**, Arthur Dehgan, Tarek Lajnef, Raphael Vallat, Jean-Baptiste Eichenlaub, Perrine Ruby, Karim Jerbi, *Unraveling the neural correlates of dream recall: Novel insights from deep convolutional nets*, Neuroinformatics, August 2018.
3. **A. Ghosh**, F. Dal Maso, M. Roig, G.D. Mitsis, M.H. Boudrias, *Hierarchical Deep Convolutional Architecture with feature visualization for analysis of EEG data*, Machine Learning for Science and Engineering (MLSE), June 2018.
4. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *AdGAP: Advanced Global Average Pooling*, AAAI-18 Student Abstract and Poster Program (SA-18), February 2018.
5. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *Training Autoencoders in Sparse Domain*, AAAI-18 Student Abstract and Poster Program (SA-18), February 2018.
6. **A. Ghosh**, F. Dal Maso, G.D. Mitsis, M.H. Boudrias, *Deep Semantic Architecture for analysis of motor task EEG data*, Montreal Artificial Intelligence & Neuroscience, November 2017. [**Facebook Best poster award**]
7. F. Dal Maso, B. Desormeau, **A. Ghosh**, M. Roig, M.H. Boudrias, *An acute bout of exercise performed immediately after motor learning alters Beta-band oscillations in the motor cortex*, The Society for Neuroscience, 47th Annual Meeting, November 2017.
8. M. Kassiopoulou, **A. Ghosh**, E. Dhamala, M.H. Boudrias, G.D. Mitsis. *Cardiac Noise Removal from BOLD fMRI based on a Dynamic Linear Model*, Organization of Human Brain Mapping (OHBM), Annual Meeting, June 2017.
9. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *Handwriting Profiling using Generative Adversarial Networks*, AAAI-17 Student Abstract and Poster Program (SA-17), February 2017. Link
10. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *SAD-GAN: Synthetic Autonomous Driving using Generative Adversarial Networks*, 30th Annual Conference on Neural Information Processing Systems (NeurIPS) 2016: Deep Learning for Action and Interaction, December 2016.Link
11. K K Agrawal, **A Ghosh**, *Generative Adversarial Motion to Image Synthesis*, 14th European Conference on Computer Vision (ECCV) 2016 Workshop on Biological and Artificial Vision, Amsterdam, October 2016.

TEACHING AND VOLUNTEERING EXPERIENCE

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| • TA for <i>Computational Perception (COMP-546)</i> , McGill University under Prof. Michael Langer. | Jan-Apr 2019 |
| • Reviewer for <i>International Conference on Learning Representations (ICLR) Reproducibility Challenge 2018</i> . | Jan 2018 |
| • Organizer for <i>Deep Learning and Neuroscience reading group</i> , Montreal Neurological Institute. | Dec 2018–Now |
| • Fellow Speaker for <i>Healthy Brains for Healthy Lives November meet 2018</i> , McGill University. | 21 Nov 2018 |
| • Volunteer for <i>Healthy Brains for Healthy Lives Symposium and Research Day 2018</i> , McGill University. | 9-10 May 2018 |
| • Tutor for <i>Deep Learning for Visual Computing Summer School 2017</i> , IIT Kharagpur under Prof. Debdeep Dey. | 2-8 Jul 2017 |

SCHOLARSHIPS AND AWARDS

- Healthy Brains for Healthy Lives (**HBHL**) Masters fellowship, September 2018 (\$10,000)
- McGill **Faculty of Medicine** Internal fellowship, September 2018 (\$10,000)
- Quebec Bio-Imaging Network (**QBIN**) travel award for Machine Learning in Science and Engineering conference at Carnegie Mellon University, Pittsburgh, USA to be held in June 2018 (\$500)
- Quebec Bio-Imaging Network (**QBIN**) Foreign Students Scholarship, December 2017 (\$7,000)
- Integrated Program in Neuroscience (**IPN**) Recruitment award 2017 (\$5,000)
- **MITACS** Graduate Fellowship award 2017 (\$15,000) for pursuing graduate studies in a Canadian university
- **MITACS** Globalink Research Internship award 2016 (\$7,500) to support the internship project during summer 2016
- **Honda** Young Engineer and Scientist award, January 2016 (\$3,000)
- **O.P. Jindal** Engineering & Management Scholarship for 2015 & 2016 (\$3,200)
- **Facebook-MAIN** Best Poster award at Montreal Artificial Intelligence & Neuroscience Conference, November 2017
- Prize winner in the 1st round (top 10) and 2nd round (top 5) at the **Dyalog APL Competition**, August 2017
- **Systems Society Award** for best project work in 8th semester on Systems among students in Electrical, Energy and Instrumentation Engineering, May 2017
- Institute **Silver medal** for securing highest CGPA among Electrical engineering graduates of IIT Kharagpur 2017

EXTRA-CURRICULAR ACTIVITIES

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|  Best All-rounder Outgoing student (2017) |  Elocution - 2 Gold, 3 Silver (2014-17) |
|  Governor , Bengali Dramatics Society (2016-17) |  Field Hockey - 1 Bronze (2016) |
|  Debate - 1 Gold, 1 Silver (2013-15) |  IOT-based Hackathon - Gold (2015) |
|  Dramatics - 10 plays, 2 Gold, 4 Silver (2013-17) |  Director , IIT Kharagpur MUN (2014) |
|  Robot Maze Traversal event - Best Freshers (2014) |  Best All-rounder fresher (2014-15) |